AM

MORTGAGE & LENDING

Payslip extraction two models must *agree* on

A production microservice for a UK mortgage broker that reads a payslip PDF, extracts every line into a strictly typed schema, and has two frontier models cross-check each other before anything reaches a human. Processing errors down 52%, manual review down 80%, and mortgage approvals 15% faster — across thousands of documents a week.

52%

FEWER PROCESSING ERRORS

80% LESS MANUAL REVIEW

15% FASTER MORTGAGE APPROVALS

1000s DOCUMENTS WEEKLY

ADAM MURPHY

AI ENGINEER · EXPERT-VETTED ON UPWORK

THE SITUATION

Every mortgage waits on someone reading a payslip

A UK mortgage broker processed thousands of payslips a week, and every one crossed a human desk. Staff read each document line by line – gross pay, tax, National Insurance, pension, the lot – and keyed the figures into the approval system by hand. Affordability decisions rested on that data, so it had to be right; but manual keying is slow and error-prone, and the review queue sat squarely in the critical path of every mortgage approval.

The brief was to take the keying out of the loop without taking the rigour out. A lending decision cannot rest on an unvalidated model guess, so "mostly right JSON" was never going to be acceptable. The service had to return data the broker's systems could trust – or say plainly that it couldn't.

WHAT I BUILT

Two models, one schema, no unchecked answers

The result is a FastAPI microservice – Dockerised, deployed on AWS – that accepts a payslip over a single endpoint and returns validated, structured JSON. Multi-page PDFs are stitched into one tall image so the vision models see the whole document at once, with nothing lost between pages. Two frontier models – GPT-4o and Gemini – extract independently and cross-validate each other, and everything lands in a Pydantic schema with 19 typed payslip line categories, so a mislabelled deduction fails loudly instead of slipping through.

01	→	02	→	03	→	04
Payslip in PDF over the API, from the broker's document platform		**Stitched** Multi-page PDFs merged into one image for the vision models		**Dual extraction** GPT-4o and Gemini cross-validate each other		**Validated JSON** Pydantic schema, 19 line categories, deterministic safety nets

Around the models sits ordinary engineering discipline: deterministic post-processing that corrects the mistakes LLMs predictably make, retries on every model call, LangSmith tracing on every request, and Slack alerts when something needs a human. Quality was measured with an evaluation harness built for the job – model output scored by the exact number of edits a human reviewer would need to reach ground truth, which maps directly onto the reviewer labour the client was paying for.

19	15%	52%	80%
TYPED PAYSLIP LINE CATEGORIES	FASTER MORTGAGE APPROVALS	FEWER PROCESSING ERRORS	LESS MANUAL REVIEW OVERHEAD

THE HARD PARTS

What it takes to trust an LLM with lending data

A mortgage cannot rest on a guess

One model reading a payslip is an opinion; two agreeing is evidence. GPT-4o and Gemini extract independently and cross-validate, with Pydantic validators enforcing the schema at the API boundary. Disagreements trigger a Slack alert rather than being quietly resolved.

"Better" measured in human edits

Accuracy percentages hide what matters, so the harness counts the edits a reviewer needs to fix each extraction – one per wrong value, more for missing lines – with an LLM-as-judge pass so "Salary" and "Basic pay" aren't marked as errors. Changes were scored against baseline before shipping.

Models make predictable mistakes

Vision models kept filing Tax and NI as deductions to sum, misreading negative adjustments and mangling name casing. A deterministic post-processing layer corrects each quirk in plain code – safety nets that hold regardless of which model sits behind them.

Architecture chosen by benchmark, not fashion

I built two full implementations – a LangChain single-chain extractor and a CrewAI multi-agent crew – and ran both through the same edit-count harness on real payslips. The evidence picked the winner; the client got the comparison, not just the conclusion.

THE RESULTS

PROCESSING ERRORS	Down **52%** – measured during the engagement, against the previous manual process
MANUAL REVIEW	Down **80%**
APPROVAL SPEED	Mortgage approvals **15%** faster
IN PRODUCTION	Thousands of documents weekly; client anonymised under NDA

WHAT THE CLIENT SAID

"Adam worked on an AI project with us and quickly demonstrated good technical skills, producing logical and well-structured code... he showed a strong commitment to delivering measurable results and consistently approached his work with a high level of professionalism. ... He would be a valuable asset to any development team."

UPWORK CLIENT REVIEW · SENIOR MACHINE LEARNING ENGINEER ENGAGEMENT · 4.9★ · $12,212.50 · 163 HOURS

ABOUT ADAM

Freelance AI engineer – **Expert-Vetted on Upwork (top 1%)**, 100% Job Success over 70+ projects, $400K+ earned, 5,750+ hours billed. I build production LLM systems for regulated industries: insurance, finance, law.

Upwork profile ↗ · github.com/codeananda ↗